

Section I: Multiple Choice

5 marks

Attempt all questions.

Answer on the multiple choice answer sheet provided for Section I.

1. In evaluating the integral $\int f(x) dx$, the substitution $t = \tan 2x$ is made.

Which of the following is the correct expression for dx ?

(A) $dx = \frac{8dt}{1+t^2}$

(B) $dx = \frac{2dt}{1+t^2}$

(C) $dx = \frac{dt}{1+t^2}$

(D) $dx = \frac{dt}{2(1+t^2)}$

2. Four students come up with the following four answers to the same indefinite integral.

Given that three of the answers are correct, which is the incorrect answer?

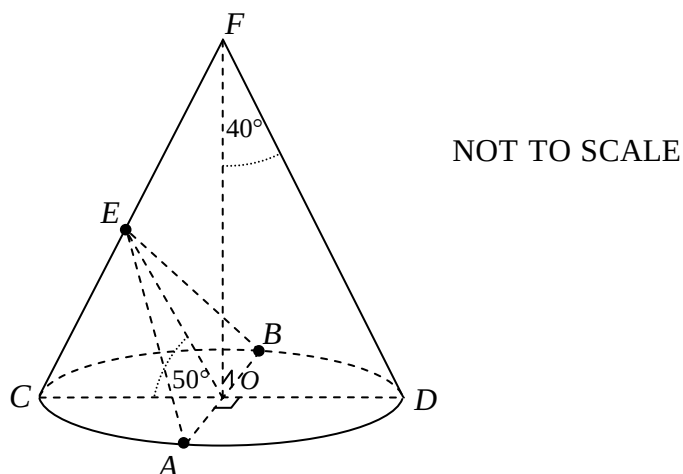
(A) $2\cos^2 x + c$

(B) $-2\sin^2 x + c$

(C) $\cos 2x + c$

(D) $-\sin 2x + c$

3. The semi-vertical angle of a hollow cone is 40° . The cone is sliced at an angle of 50° to the base, so that the slice includes the triangle ABE .



What is the shape of the cross-section formed?

(A) Circle

(B) Ellipse

(C) Hyperbola

(D) Parabola

4. The equation $|x-2|+|x+2|=6$ corresponds to an ellipse in the Argand diagram. What is the length of the semi-minor axis of this ellipse?

(A) 3 (B) $\sqrt{5}$
(C) $\sqrt{13}$ (D) 6

5. A conic curve (either an ellipse or a hyperbola) is centred on the origin. A focus is located at $(4,0)$, one of its directrices is $x=9$ and the point $(-6,0)$ lies on the curve.

What is the eccentricity of this conic?

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$
(C) $\frac{3}{2}$ (D) 2

Section II

Attempt all questions.

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (14 marks)

(a) Find $\int 2x\sqrt{2x-3} \, dx$. 3

(b) (i) Write $\frac{8}{(x+2)(x^2+4)}$ in the form $\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$ 2

(ii) Hence evaluate $\int_0^2 \frac{8 \, dx}{(x+2)(x^2+4)}$ 3

(c) Find $\int \frac{dx}{(4-x^2)^{\frac{3}{2}}}$ 3

(d) Evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$ 3

Question 7 (14 marks)

(a) Consider the ellipse $12x^2 + 16y^2 = 192$.

- (i) Find the eccentricity of this ellipse. 1
- (ii) Draw a neat sketch of this ellipse, showing the coordinates of the vertices and the foci, and the equations of the directrices. 3
- (iii) Derive the equation of the tangent to the ellipse at the point $P(2, 3)$. 2
- (iv) Consider the triangle PBS' , where B is the y intercept of this tangent and S' is the focus furthest from P . Find the area of this triangle. 2

(b) $P\left(2p, \frac{2}{p}\right)$ is a point on the hyperbola $xy = 4$.

- (i) Show that the equation of the normal at P is $y = p^2x - 2p^3 + \frac{2}{p}$. 3
- (ii) Find the coordinates of the midpoint M of PQ , where Q is the point where the normal at P meets the x -axis. 2
- (iii) Hence find the equation of the locus of M as P moves on the hyperbola. 1

Question 8 (14 marks)

- (a) (i) Show that the equation of the tangent to the hyperbola $x^2 - y^2 = 4$ at the point $P(2\sec\theta, 2\tan\theta)$ is $x\sec\theta - y\tan\theta = 2$. 2

- (ii) Show that this tangent intersects the asymptotes of the hyperbola at the points 2

$$A\left(\frac{2\cos\theta}{1-\sin\theta}, \frac{2\cos\theta}{1-\sin\theta}\right) \text{ and } B\left(\frac{2\cos\theta}{1+\sin\theta}, \frac{-2\cos\theta}{1+\sin\theta}\right).$$

- (iii) Hence find the ratio $PA:PB$ as a number that is independent of θ . 3

- (b) Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, where $n \geq 2$.

- (i) Show that $I_n = \frac{n-1}{n} I_{n-2}$. 3

- (ii) Show that for any continuous function $f(x)$, $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. 1

- (iii) By first considering $\int_0^{\frac{\pi}{2}} x \sin^6 x \, dx$, use parts (i) and (ii) to evaluate 3

$$\int_0^{\frac{\pi}{2}} x(\sin^6 x + \cos^6 x) \, dx.$$

End of paper

Extension 2 Assessment 3 2013 – Solutions

Section I

1. D

$$t = \tan 2x$$

$$x = \frac{1}{2} \tan^{-1} t$$

$$dx = \frac{dt}{2(1+t^2)}$$

2. D

$$\frac{d}{dx}(2 \cos^2 x) = 4 \cos x \cdot (-\sin x) = -4 \sin x \cos x = -2 \sin 2x$$

$$\frac{d}{dx}(-2 \sin^2 x) = -4 \sin x \cdot \cos x = -4 \sin x \cos x = -2 \sin 2x$$

$$\frac{d}{dx}(\cos 2x) = -2 \sin 2x$$

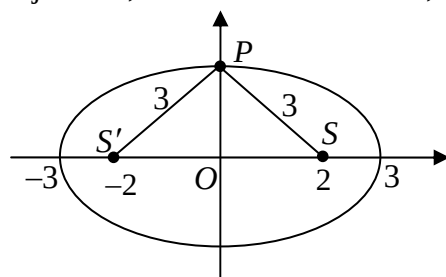
$$\frac{d}{dx}(-\sin 2x) = -2 \cos 2x$$

3. D

Through simple geometrical arguments, $OE \parallel DF$, so it is a parabola.

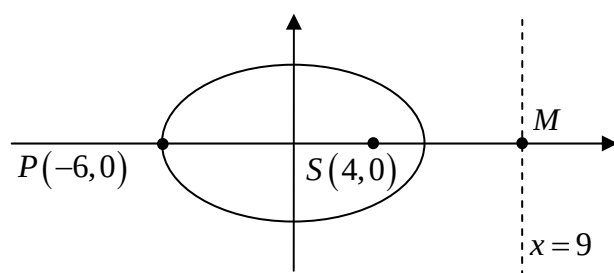
4. B

The equation corresponds to $PS + PS' = 2a$ (property of an ellipse) where a is the semi-major axis, S and S' are the foci, and P is an arbitrary point on the ellipse.



Using Pythagoras', $OP = \sqrt{5}$

5. B



$$\begin{aligned}
 e &= \frac{PS}{PM} \\
 &= \frac{10}{15} \\
 &= \frac{2}{3}
 \end{aligned}$$

Section II

Question 6

(a) Find $\int 2x\sqrt{2x-3} dx$.

3

$$\begin{aligned}\int 2x\sqrt{2x-3} dx &= \int (u+3) \cdot u^{\frac{1}{2}} \cdot \frac{1}{2} du && \text{Let } u = 2x-3 \Rightarrow 2x = u+3 \\ &= \frac{1}{2} \int \left(u^{\frac{3}{2}} + 3u^{\frac{1}{2}} \right) du && du = 2 dx \Rightarrow dx = \frac{1}{2} du \\ &= \frac{1}{2} \left(\frac{2}{5} u^{\frac{5}{2}} + 2u^{\frac{3}{2}} \right) + c \\ &= \frac{1}{5} (2x-3)^{\frac{5}{2}} + (2x-3)^{\frac{3}{2}} + c\end{aligned}$$

OR

$$\begin{aligned}\int 2x\sqrt{2x-3} dx &= \int (u^2+3) \cdot u \cdot u du && \text{Let } u^2 = 2x-3 \Rightarrow 2x = u^2+3 \\ &= \int (u^4 + 3u^2) du && 2u du = 2 dx \Rightarrow dx = u du \\ &= \frac{1}{5} u^5 + u^3 + c \\ &= \frac{1}{5} (2x-3)^{\frac{5}{2}} + (2x-3)^{\frac{3}{2}} + c\end{aligned}$$

(b) (i) Write $\frac{8}{(x+2)(x^2+4)}$ in the form $\frac{A}{x+2} + \frac{Bx+C}{x^2+4}$

2

$$\begin{aligned}\text{Let } \frac{8}{(x+2)(x^2+4)} &= \frac{A}{x+2} + \frac{Bx+C}{x^2+4} \\ 8 &= A(x^2+4) + (Bx+C)(x+2) \\ (x=-2) \quad 8 &= 8A \quad \Rightarrow \quad A=1 \\ (x=0) \quad 8 &= 4A+2C \quad \Rightarrow \quad 8=4+2C \quad \Rightarrow \quad C=2 \\ (x=1) \quad 8 &= 5A+3B+3C \quad \Rightarrow \quad 8=5+3B+6 \quad \Rightarrow \quad B=-1 \\ \therefore \frac{8}{(x+2)(x^2+4)} &= \frac{1}{x+2} + \frac{-x+2}{x^2+4}\end{aligned}$$

(ii) Hence evaluate $\int_0^2 \frac{8dx}{(x+2)(x^2+4)}$

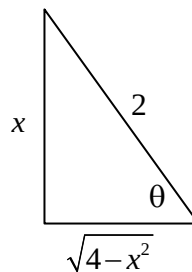
3

$$\begin{aligned}\int_0^2 \frac{8dx}{(x+2)(x^2+4)} &= \int_0^2 \left(\frac{1}{x+2} - \frac{x}{x^2+4} + \frac{2}{x^2+4} \right) dx \\ &= \left[\ln(x+2) - \frac{1}{2} \ln(x^2+4) + \tan^{-1} \frac{x}{2} \right]_0^2 \\ &= \left(\ln 4 - \frac{1}{2} \ln 8 + \frac{\pi}{4} \right) - \left(\ln 2 - \frac{1}{2} \ln 4 + 0 \right) \\ &= 2 \ln 2 - \frac{3}{2} \ln 2 + \frac{\pi}{4} - \ln 2 + \ln 2 \\ &= \frac{\pi}{4} + \frac{1}{2} \ln 2\end{aligned}$$

(c) Find $\int \frac{dx}{(4-x^2)^{\frac{3}{2}}}$

3

$$\begin{aligned}\int \frac{dx}{(4-x^2)^{\frac{3}{2}}} &= \int \frac{2 \cos \theta d\theta}{(4-4 \sin^2 \theta)^{\frac{3}{2}}} \quad \text{Let } x = 2 \sin \theta \\ & \quad dx = 2 \cos \theta d\theta \\ &= \int \frac{2 \cos \theta d\theta}{8 \cos^3 \theta} \\ &= \frac{1}{4} \int \sec^2 \theta d\theta \\ &= \frac{1}{4} \tan \theta + c \\ &= \frac{x}{4\sqrt{4-x^2}} + c\end{aligned}$$



OR

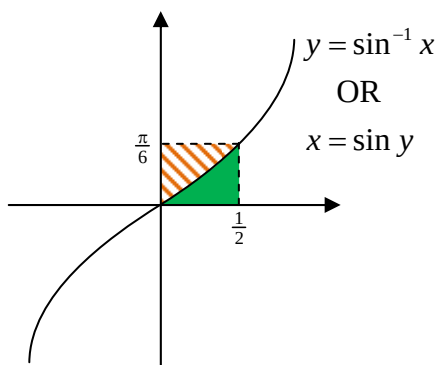
$$\begin{aligned}\int (4-x^2)^{-\frac{1}{2}} dx &= \int (4-x^2)^{-\frac{1}{2}} \cdot \frac{d}{dx}(x) dx \\ &= x(4-x^2)^{-\frac{1}{2}} - \int x \cdot \frac{1}{2} (4-x^2)^{-\frac{3}{2}} \cdot (-2x) dx \\ &= \frac{x}{\sqrt{4-x^2}} - \int x^2 (4-x^2)^{-\frac{3}{2}} dx \\ &= \frac{x}{\sqrt{4-x^2}} - \int [4 - (4-x^2)] \cdot (4-x^2)^{-\frac{3}{2}} dx \\ \int (4-x^2)^{-\frac{1}{2}} dx &= \frac{x}{\sqrt{4-x^2}} - 4 \int (4-x^2)^{-\frac{3}{2}} dx + \int (4-x^2)^{-\frac{1}{2}} dx \\ \int \frac{dx}{(4-x^2)^{\frac{3}{2}}} &= \frac{x}{4\sqrt{4-x^2}} + c\end{aligned}$$

(d) Evaluate $\int_0^{\frac{1}{2}} \sin^{-1} x \, dx$

3

$$\begin{aligned}
 \int_0^{\frac{1}{2}} \sin^{-1} x \, dx &= \int_0^{\frac{1}{2}} \sin^{-1} x \cdot \frac{d}{dx}(x) \cdot dx \\
 &= \left[x \sin^{-1} x \right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} x \cdot \frac{d}{dx}(\sin^{-1} x) \cdot dx \\
 &= \frac{1}{2} \cdot \frac{\pi}{6} - 0 - \int_0^{\frac{1}{2}} x \cdot \frac{1}{\sqrt{1-x^2}} \cdot dx \\
 &= \frac{\pi}{12} + \frac{1}{2} \int_0^{\frac{1}{2}} (1-x^2)^{-\frac{1}{2}} \cdot (-2x \, dx) \\
 &= \frac{\pi}{12} + \frac{1}{2} \cdot \left[\frac{(1-x^2)^{\frac{1}{2}}}{\cancel{1/2}} \right]_0^{\frac{1}{2}} \\
 &= \frac{\pi}{12} + \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}} \\
 &= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1
 \end{aligned}$$

OR



$$\begin{aligned}
 \int_0^{\frac{1}{2}} \sin^{-1} x \, dx &= \text{Area of solid region} \\
 &= \text{Area of rectangle} - \text{Area of hashed region} \\
 &= \frac{1}{2} \cdot \frac{\pi}{6} - \int_0^{\frac{\pi}{6}} \sin y \, dy \\
 &= \frac{\pi}{12} + [\cos y]_0^{\frac{\pi}{6}} \\
 &= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1
 \end{aligned}$$

Question 7

(a) Consider the ellipse $12x^2 + 16y^2 = 192$.

(i) Find the eccentricity of this ellipse.

1

$$12x^2 + 16y^2 = 192$$

$$(\div 192) \quad \frac{x^2}{16} + \frac{y^2}{12} = 1$$

$$b^2 = a^2(1 - e^2)$$

$$12 = 16(1 - e^2)$$

$$1 - e^2 = \frac{3}{4}$$

$$e^2 = \frac{1}{4}$$

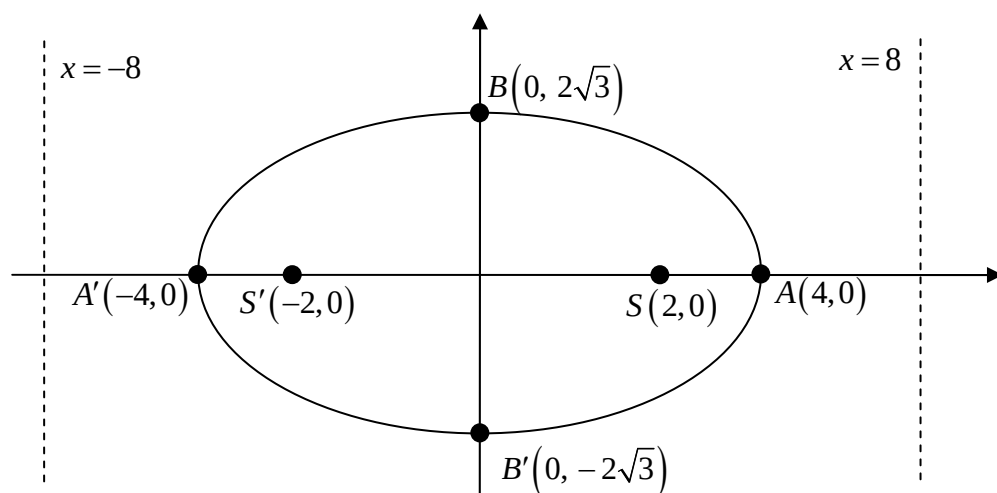
$$e = \frac{1}{2}$$

(ii) Draw a neat sketch of this ellipse, showing the coordinates of the vertices and the foci, and the equations of the directrices.

3

$$ae = 4 \times \frac{1}{2} = 2 \quad \Rightarrow \quad S(\pm 2, 0)$$

$$\frac{a}{e} = \frac{4}{1/2} = 8 \quad \Rightarrow \quad \text{directrices: } x = \pm 8$$



(iii) Derive the equation of the tangent to the ellipse at the point $P(2, 3)$.

2

$$12x^2 + 16y^2 = 192$$

$$24x + 32y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{3x}{4y}$$

$$\text{At } P(2, 3), \quad m_T = -\frac{1}{2}$$

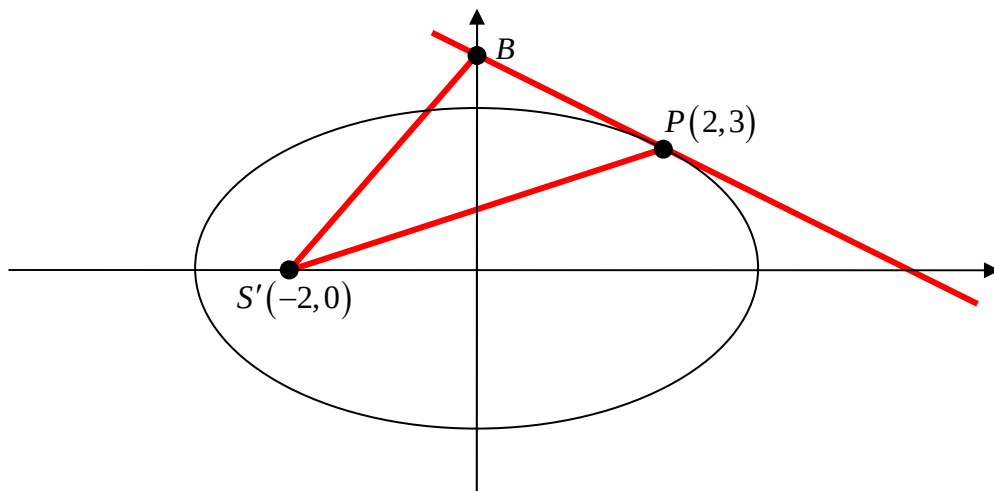
$$\text{Tangent: } y - 3 = -\frac{1}{2}(x - 2)$$

$$2y - 6 = -x + 2$$

$$\mathbf{x + 2y - 8 = 0}$$

(iv) The vertices of a triangle are P ; B , the y intercept of this tangent; and S' , the focus furthest from P . Find the area of this triangle.

3



$$B(0, 4)$$

$$\therefore PB = \sqrt{5}$$

$$\begin{aligned} \text{Perpendicular distance from } S' \text{ to } PB: \quad d &= \frac{|1(-2) + 2(0) - 8|}{\sqrt{1^2 + 2^2}} \\ &= \frac{10}{\sqrt{5}} \end{aligned}$$

$$\therefore \text{Area } \triangle PBS' = \frac{1}{2} \times \sqrt{5} \times \frac{10}{\sqrt{5}}$$

$$\mathbf{\text{Area} = 5 \text{ units}^2}$$

(b) $P\left(2p, \frac{2}{p}\right)$ is a point on the hyperbola $xy = 4$.

(i) Show that the equation of the normal at P is $y = p^2x - 2p^3 + \frac{2}{p}$.

3

$$\begin{aligned} x &= 2p & y &= \frac{2}{p} \\ \frac{dx}{dp} &= 2 & \frac{dy}{dp} &= -\frac{2}{p^2} \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{dp} \cdot \frac{dp}{dx} \\ &= -\frac{2}{p^2} \cdot 2 \\ &= -\frac{4}{p^2} \\ \therefore m_N &= p^2 \end{aligned}$$

$$\begin{aligned} \text{Normal: } y - \frac{2}{p} &= p^2(x - 2p) \\ &= p^2x - 2p^3 \\ y &= p^2x - 2p^3 + \frac{2}{p} \end{aligned}$$

(ii) Find the coordinates of the midpoint M of PQ , where Q is the point where the normal at P meets the x -axis.

2

$$\begin{aligned} x\text{-intercepts } (y=0): \quad p^2x &= 2p^3 - \frac{2}{p} \\ x &= 2p - \frac{2}{p^3} \end{aligned}$$

$$\begin{aligned} \therefore Q\left(2p - \frac{2}{p^3}, 0\right), \quad P\left(2p, \frac{2}{p}\right) \\ M\left(\frac{1}{2}\left[2p + \left(2p - \frac{2}{p^3}\right)\right], \frac{1}{2}\left(\frac{2}{p} + 0\right)\right) \Rightarrow M\left(2p - \frac{1}{p^3}, \frac{1}{p}\right) \end{aligned}$$

(iii) Hence find the equation of the locus of M as P moves on the hyperbola.

1

$$\begin{aligned} y &= \frac{1}{p} \Rightarrow p = \frac{1}{y} \\ x &= 2p - \frac{1}{p^3} \\ x &= \frac{2}{y} - y^3 \end{aligned}$$

Question 8

- (a) (i) Show that the equation of the tangent to the hyperbola $x^2 - y^2 = 4$ at the point $P(2 \sec \theta, 2 \tan \theta)$ is $x \sec \theta - y \tan \theta = 2$. 2

$$\begin{aligned} x &= 2 \sec \theta & y &= 2 \tan \theta \\ \frac{dx}{d\theta} &= 2 \sec \theta \tan \theta & \frac{dy}{d\theta} &= 2 \sec^2 \theta \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} \\ &= \frac{2 \sec^2 \theta}{2 \sec \theta \tan \theta} \\ &= \frac{\sec \theta}{\tan \theta} \end{aligned}$$

$$\begin{aligned} \text{Tangent: } y - 2 \tan \theta &= \frac{\sec \theta}{\tan \theta} (x - 2 \sec \theta) \\ y \tan \theta - 2 \tan^2 \theta &= x \sec \theta - 2 \sec^2 \theta \\ x \sec \theta - y \tan \theta &= 2 (\sec^2 \theta - \tan^2 \theta) \\ x \sec \theta - y \tan \theta &= 2 \end{aligned}$$

- (ii) Show that this tangent intersects the asymptotes of the hyperbola at the points 2

$$A\left(\frac{2 \cos \theta}{1 - \sin \theta}, \frac{2 \cos \theta}{1 - \sin \theta}\right) \text{ and } B\left(\frac{2 \cos \theta}{1 + \sin \theta}, \frac{-2 \cos \theta}{1 + \sin \theta}\right).$$

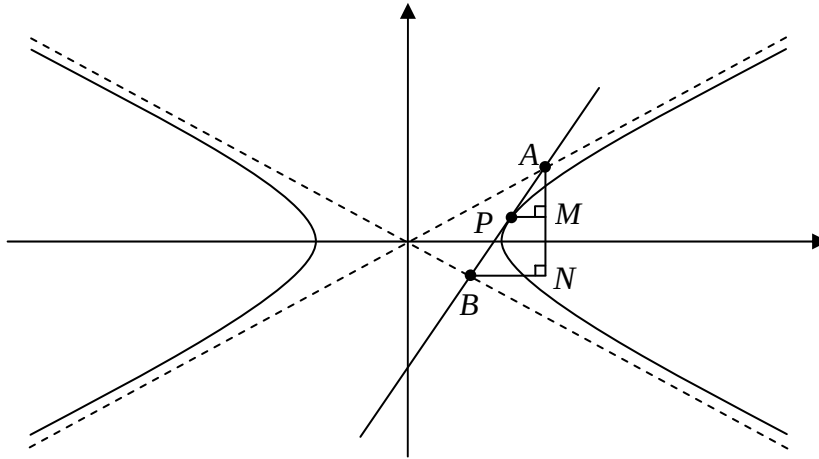
$$\text{Asymptotes: } y = \pm x$$

$$x \sec \theta - (\pm x) \tan \theta = 2$$

$$x(\sec \theta \mp \tan \theta) = 2$$

$$\begin{aligned} x &= \frac{2}{\sec \theta \mp \tan \theta} \times \frac{\cos \theta}{\cos \theta} \\ &= \frac{2 \cos \theta}{1 \mp \sin \theta} \\ y &= \pm \frac{2 \cos \theta}{1 \mp \sin \theta} \end{aligned}$$

$$\text{ie. } A\left(\frac{2 \cos \theta}{1 - \sin \theta}, \frac{2 \cos \theta}{1 - \sin \theta}\right), B\left(\frac{2 \cos \theta}{1 + \sin \theta}, \frac{-2 \cos \theta}{1 + \sin \theta}\right)$$



$$\frac{PA}{PB} = \frac{MA}{MN} \quad (\text{parallel lines preserve ratio})$$

$$\begin{aligned}
 &= \frac{\frac{\cos \theta}{1 - \sin \theta} - \tan \theta}{\tan \theta + \frac{\cos \theta}{1 + \sin \theta}} \times \frac{(1 - \sin \theta)(1 + \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)} \\
 &= \frac{(1 + \sin \theta)[\cos \theta - \tan \theta(1 - \sin \theta)]}{(1 - \sin \theta)[\tan \theta(1 + \sin \theta) + \cos \theta]} \times \frac{\cos \theta}{\cos \theta} \\
 &= \frac{(1 + \sin \theta)(\cos^2 \theta - \sin \theta + \sin^2 \theta)}{(1 - \sin \theta)(\sin \theta + \sin^2 \theta + \cos^2 \theta)} \\
 &= \frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)} \\
 &= 1
 \end{aligned}$$

ie. $PA:PB = 1:1$

(b) Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, where $n \geq 2$.

(i) Show that $I_n = \frac{n-1}{n} I_{n-2}$.

3

$$\begin{aligned}
 I_n &= \int_0^{\frac{\pi}{2}} \cos^n x \, dx \\
 &= \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cdot \cos x \cdot dx \\
 &= \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cdot d(\sin x) \\
 &= \left[\cos^{n-1} x \cdot \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \cdot d(\cos^{n-1} x) \\
 &= 0 - 0 - \int_0^{\frac{\pi}{2}} \sin x \cdot (n-1) \cos^{n-2} x \cdot (-\sin x) \cdot dx \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \cdot \sin^2 x \cdot dx \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \cdot (1 - \cos^2 x) \cdot dx \\
 &= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \, dx - (n-1) \int_0^{\frac{\pi}{2}} \cos^n x \, dx \\
 &= (n-1) I_{n-1} - (n-1) I_n \\
 [1 + (n-1)] I_n &= (n-1) I_{n-1} \\
 I_n &= \frac{n-1}{n} I_{n-1}
 \end{aligned}$$

(ii) Show that for any continuous function $f(x)$, $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$.

1

$$\begin{aligned}
 \int_0^a f(x) \, dx &= \int_a^0 f(a-u) \cdot (-du) && \text{Let } x = a - u \\
 &= \int_0^a f(a-u) \, du && dx = -du \\
 &= \int_0^a f(a-x) \, dx && (x=0) \quad u=a \\
 &&& (x=a) \quad u=0 \\
 &&& \text{(indef integ indept of choice of var)}
 \end{aligned}$$

(iii) By first considering $\int_0^{\frac{\pi}{2}} x \sin^6 x \, dx$, use parts (i) and (ii) to evaluate

3

$$\int_0^{\frac{\pi}{2}} x (\sin^6 x + \cos^6 x) \, dx.$$

$$\int_0^{\frac{\pi}{2}} x \sin^6 x \, dx = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \sin^6 \left(\frac{\pi}{2} - x \right) \, dx \quad (\text{from part ii})$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x \right) \cos^6 x \, dx$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \cos^6 x \, dx - \int_0^{\frac{\pi}{2}} x \cos^6 x \, dx$$

$$\int_0^{\frac{\pi}{2}} x \sin^6 x \, dx + \int_0^{\frac{\pi}{2}} x \cos^6 x \, dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \cos^6 x \, dx$$

$$\int_0^{\frac{\pi}{2}} x (\sin^6 x + \cos^6 x) \, dx = \frac{\pi}{2} I_6 \quad (\text{where } I_n \text{ is defined as in part i})$$

$$I_2 = \int_0^{\frac{\pi}{2}} \cos^2 x \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos 2x) \, dx$$

$$= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left(\frac{\pi}{2} + 0 - 0 - 0 \right)$$

$$= \frac{\pi}{4}$$

$$I_4 = \frac{3}{4} I_2 = \frac{3\pi}{16}$$

$$I_6 = \frac{5}{6} I_4 = \frac{15\pi}{96}$$

$$\therefore \int_0^{\frac{\pi}{2}} x (\sin^6 x + \cos^6 x) \, dx = \frac{15\pi^2}{192}$$